

Agentic-AI

Agent Frameworks: AutoGen & CrewAI

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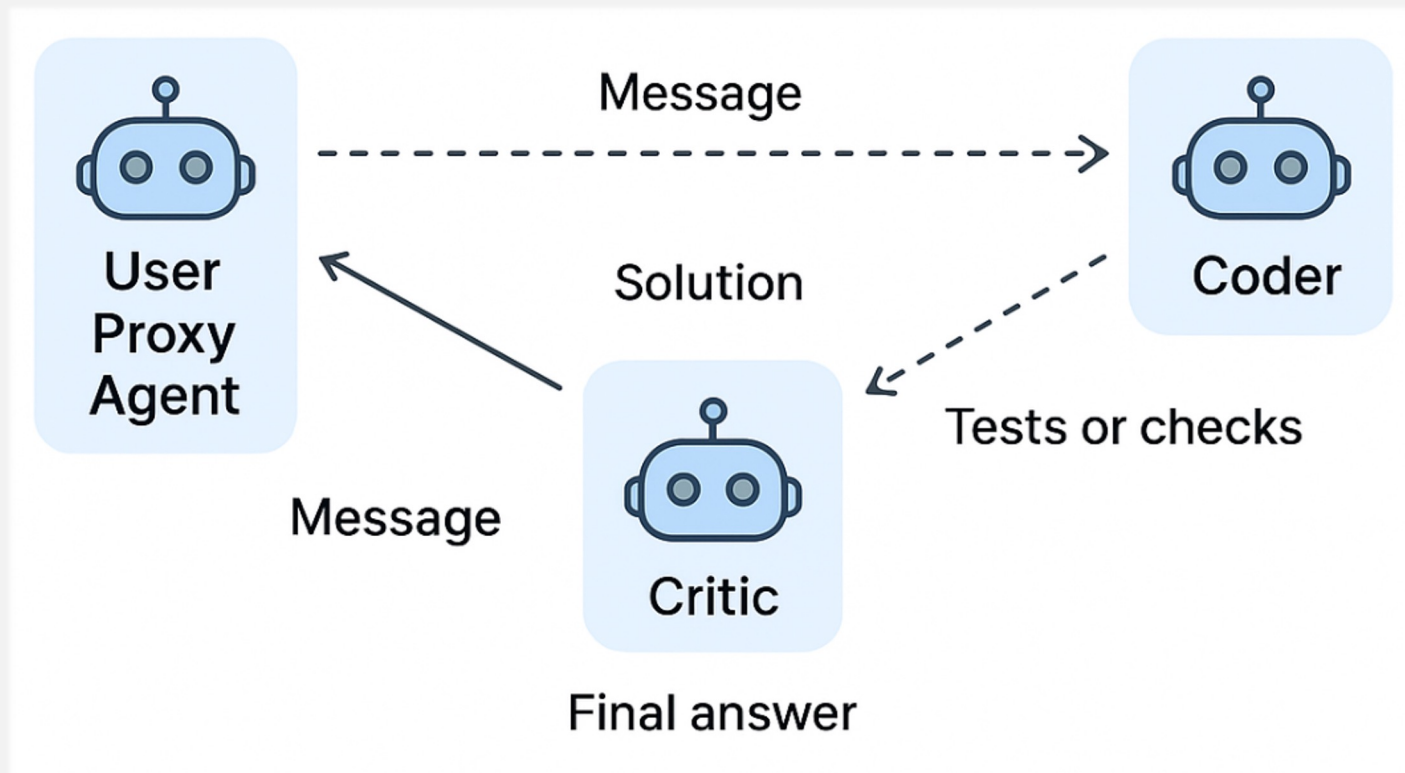


Bundesministerium
für Forschung, Technologie
und Raumfahrt



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AutoGen



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AutoGen

Sequential Chat Example

Part 1

```
In [ ]: onboarding_personal_information_agent = ConversableAgent(
        name="Onboarding Personal Information Agent",
        system_message='''You are a helpful customer onboarding agent,
        you are here to help new customers get started with our product.
        Your job is to gather customer's name and location.
        Do not ask for other information. Return 'TERMINATE'
        when you have gathered all the information.''' ,
        llm_config=llm_config,
        code_execution_config=False,
        human_input_mode="NEVER",
    )
```

```
In [ ]: onboarding_topic_preference_agent = ConversableAgent(
        name="Onboarding Topic preference Agent",
        system_message='''You are a helpful customer onboarding agent,
        you are here to help new customers get started with our product.
        Your job is to gather customer's preferences on news topics.
        Do not ask for other information.
        Return 'TERMINATE' when you have gathered all the information.''' ,
        llm_config=llm_config,
        code_execution_config=False,
        human_input_mode="NEVER",
    )
```

```
In [ ]: customer_engagement_agent = ConversableAgent(
        name="Customer Engagement Agent",
        system_message='''You are a helpful customer service agent
        here to provide fun for the customer based on the user's
        personal information and topic preferences.
        This could include fun facts, jokes, or interesting stories.
        Make sure to make it engaging and fun!
        Return 'TERMINATE' when you are done.''' ,
        llm_config=llm_config,
        code_execution_config=False,
        human_input_mode="NEVER",
        is_termination_msg=lambda msg: "terminate" in msg.get("content").lower(),
    )
```

```
In [ ]: customer_proxy_agent = ConversableAgent(
        name="customer_proxy_agent",
        llm_config=False,
        code_execution_config=False,
        human_input_mode="ALWAYS",
        is_termination_msg=lambda msg: "terminate" in msg.get("content").lower(),
    )
```

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Ref: <https://learn.deeplearning.ai/courses/ai-agentic-design-patterns-with-autogen/lesson/hsm3a/sequential-chats-and-customer-onboarding>

AutoGen

Sequential Chat Example

Part 2

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Creating tasks

Now, you can craft a series of tasks to facilitate the onboarding process.

```
In [7]: chats = [
    {
        "sender": onboarding_personal_information_agent,
        "recipient": customer_proxy_agent,
        "message":
            "Hello, I'm here to help you get started with our product."
            "Could you tell me your name and location?",
        "summary_method": "reflection_with_llm",
        "summary_args": {
            "summary_prompt": "Return the customer information "
                               "into as JSON object only: "
                               "'{'name': '', 'location': ''}'",
        },
        "max_turns": 2,
        "clear_history": True
    },
    {
        "sender": onboarding_topic_preference_agent,
        "recipient": customer_proxy_agent,
        "message":
            "Great! Could you tell me what topics you are "
            "interested in reading about?",
        "summary_method": "reflection_with_llm",
        "max_turns": 1,
        "clear_history": False
    },
    {
        "sender": customer_proxy_agent,
        "recipient": customer_engagement_agent,
        "message": "Let's find something fun to read.",
        "max_turns": 1,
        "summary_method": "reflection_with_llm",
    },
]
```

Start the onboarding process

Note: You might get a slightly different response than what's shown in the video. Feel free to try different inputs, such as name, location, and preferences.

```
In [*]: from autogen import initiate_chats

        chat_results = initiate_chats(chats)
```

```
*****
Starting a new chat...
*****
Onboarding Personal Information Agent (to customer_proxy_agent):
```

Hello, I'm here to help you get started with our product.Could you tell me your name an

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Ref: <https://learn.deeplearning.ai/courses/ai-agentic-design-patterns-with-autogen/lesson/hsm3a/sequential-chats-and-customer-onboarding>

Code

Writing & Execution

Setup

```
In [1]: llm_config = {"model": "gpt-4-turbo"}
```

Define a code executor

```
In [2]: from autogen.coding import LocalCommandLineCodeExecutor
```

```
In [3]: executor = LocalCommandLineCodeExecutor(  
    timeout=60,  
    work_dir="coding",  
)
```

Create agents

```
In [4]: from autogen import ConversableAgent, AssistantAgent
```

1. Agent with code executor configuration

```
In [5]: code_executor_agent = ConversableAgent(  
    name="code_executor_agent",  
    llm_config=False,  
    code_execution_config={"executor": executor},  
    human_input_mode="ALWAYS",  
    default_auto_reply=  
        "Please continue. If everything is done, reply 'TERMINATE'.",  
)
```

2. Agent with code writing capability

```
In [6]: code_writer_agent = AssistantAgent(  
    name="code_writer_agent",  
    llm_config=llm_config,  
    code_execution_config=False,  
    human_input_mode="NEVER",  
)
```

```
In [11]: code_writer_agent_system_message = code_writer_agent.system_message
```

```
In [12]: print(code_writer_agent_system_message)
```

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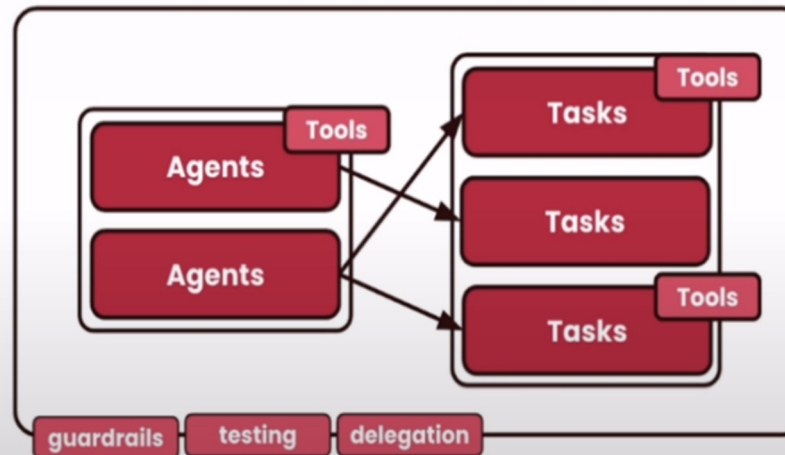
Ref: <https://learn.deeplearning.ai/courses/ai-agentic-design-patterns-with-autogen/lesson/dxvho/coding-and-financial-analysis>

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CrewAI

Crew



Agents

- Role
- Goal
- Backstory

```
researcher:  
role: >  
  {topic} Senior Data Researcher  
goal: >  
  Uncover cutting-edge developments in {topic}  
backstory: >  
  You're a seasoned researcher with a knack for uncovering...
```

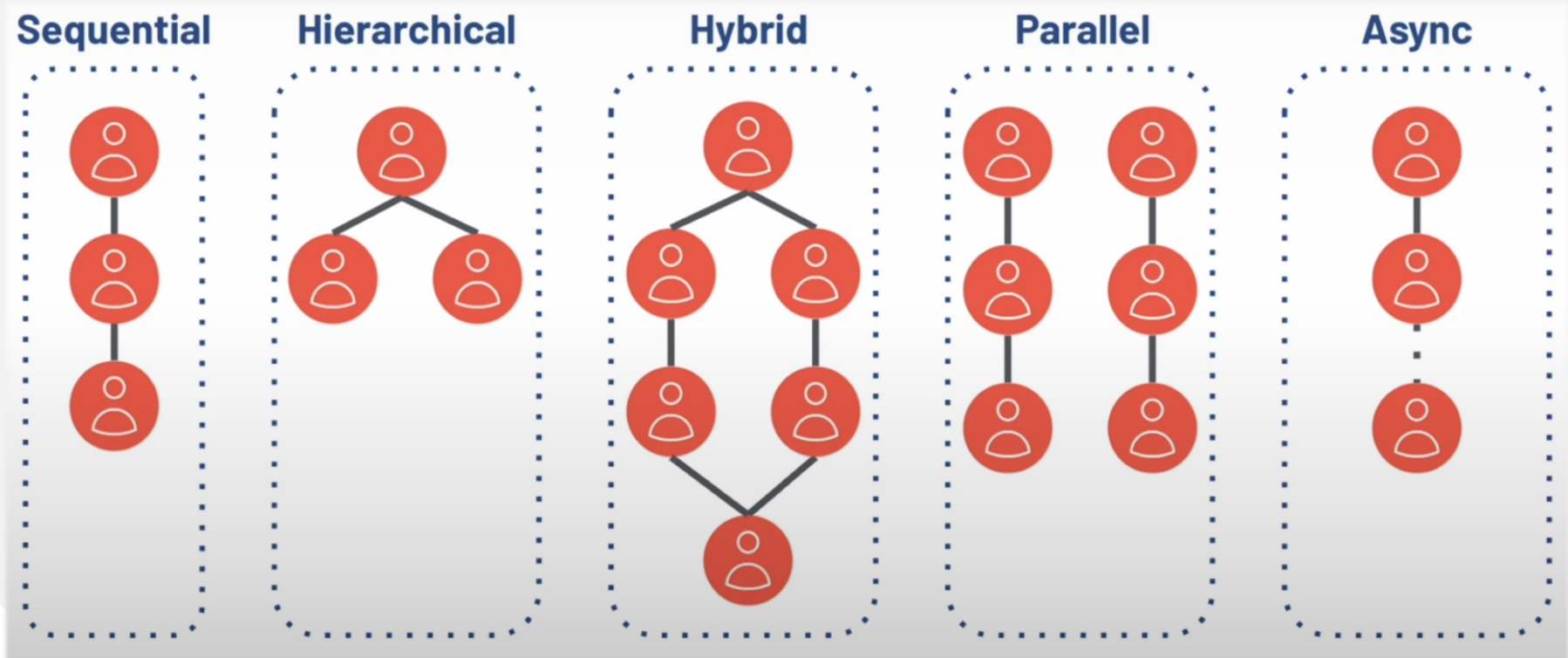
Tasks

- Description
- Expected Output
- Agent

```
research_task:  
description: >  
  Conduct a thorough research about {topic}...  
expected_output: >  
  10 points of the most relevant information about {topic}...  
agent: researcher
```

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Process Types



Gefördert durch:

agents.yaml

agents.yaml

```
# src/latest_ai_development/config/agents.yaml
researcher:
  role: >
    {topic} Senior Data Researcher
  goal: >
    Uncover cutting-edge developments in {topic}
  backstory: >
    You're a seasoned researcher with a knack for uncovering the latest
    developments in {topic}. Known for your ability to find the most relevant
    information and present it in a clear and concise manner.

reporting_analyst:
  role: >
    {topic} Reporting Analyst
  goal: >
    Create detailed reports based on {topic} data analysis and research findings
  backstory: >
    You're a meticulous analyst with a keen eye for detail. You're known for
    your ability to turn complex data into clear and concise reports, making
    it easy for others to understand and act on the information you provide.
```

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Ref: <https://docs.crewai.com/en/quickstart>

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tasks.yaml

tasks.yaml

```
# src/latest_ai_development/config/tasks.yaml
research_task:
  description: >
    Conduct a thorough research about {topic}
    Make sure you find any interesting and relevant information given
    the current year is 2025.
  expected_output: >
    A list with 10 bullet points of the most relevant information about {topic}
  agent: researcher

reporting_task:
  description: >
    Review the context you got and expand each topic into a full section for a report.
    Make sure the report is detailed and contains any and all relevant information.
  expected_output: >
    A fully fledged reports with the main topics, each with a full section of information.
    Formatted as markdown without '```'
  agent: reporting_analyst
  output_file: report.md
```

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crew.py

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Ref: <https://docs.crewai.com/en/quickstart>

crew.py

```
# src/latest_ai_development/crew.py
from crewai import Agent, Crew, Process, Task
from crewai.project import CrewBase, agent, crew, task
from crewai_tools import SerperDevTool
from crewai.agents.agent_builder.base_agent import BaseAgent
from typing import List

@CrewBase
class LatestAiDevelopmentCrew():
    """LatestAiDevelopment crew"""

    agents: List[BaseAgent]
    tasks: List[Task]

    @agent
    def researcher(self) -> Agent:
        return Agent(
            config=self.agents_config['researcher'], # type: ignore[index]
            verbose=True,
            tools=[SerperDevTool()]
        )

    @agent
    def reporting_analyst(self) -> Agent:
        return Agent(
            config=self.agents_config['reporting_analyst'], # type: ignore[index]
            verbose=True
        )

    @task
    def research_task(self) -> Task:
        return Task(
            config=self.tasks_config['research_task'], # type: ignore[index]
        )

    @task
    def reporting_task(self) -> Task:
        return Task(
            config=self.tasks_config['reporting_task'], # type: ignore[index]
            output_file='output/report.md' # This is the file that will contain the final report.
        )

    @crew
    def crew(self) -> Crew:
        """Creates the LatestAiDevelopment crew"""
        return Crew(
            agents=self.agents, # Automatically created by the @agent decorator
            tasks=self.tasks, # Automatically created by the @task decorator
            process=Process.sequential,
            verbose=True,
        )
```

main.py

main.py

```
#!/usr/bin/env python
# src/latest_ai_development/main.py
import sys
from latest_ai_development.crew import LatestAiDevelopmentCrew

def run():
    """
    Run the crew.
    """
    inputs = {
        'topic': 'AI Agents'
    }
    LatestAiDevelopmentCrew().crew().kickoff(inputs=inputs)
```

- Lock the dependencies and install them by using the CLI command:

Terminal

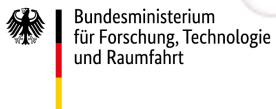
```
crewai install
```

- To run your crew, execute the following command in the root of your project:

Terminal

```
crewai run
```

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Ref: <https://docs.crewai.com/en/quickstart>

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report.md

Here's an example of what the report should look like:

```
output/report.md

# Comprehensive Report on the Rise and Impact of AI Agents in 2025

## 1. Introduction to AI Agents
In 2025, Artificial Intelligence (AI) agents are at the forefront of innovation across various industries. As int

## 2. Benefits of AI Agents
AI agents bring numerous advantages that are transforming traditional work environments. Key benefits include:

- **Task Automation**: AI agents can carry out repetitive tasks such as data entry, scheduling, and payroll proces
- **Improved Efficiency**: By quickly processing large datasets and performing analyses that would take humans si
- **Enhanced Decision-Making**: AI agents can analyze trends and patterns in data, provide insights, and even sug

## 3. Popular AI Agent Frameworks
Several frameworks have emerged to facilitate the development of AI agents, each with its own unique features and

- **Autogen**: A framework designed to streamline the development of AI agents through automation of code generat
- **Semantic Kernel**: Focuses on natural language processing and understanding, enabling agents to comprehend us
- **Promptflow**: Provides tools for developers to create conversational agents that can navigate complex interac
- **Langchain**: Specializes in leveraging various APIs to ensure agents can access and utilize external data eff
- **CrewAI**: Aimed at collaborative environments, CrewAI strengthens teamwork by facilitating communication thro
- **MemGPT**: Combines memory-optimized architectures with generative capabilities, allowing for more personalize

These frameworks empower developers to build versatile and intelligent agents that can engage users, perform adva

## 4. AI Agents in Human Resources
AI agents are revolutionizing HR practices by automating and optimizing key functions:

- **Recruiting**: AI agents can screen resumes, schedule interviews, and even conduct initial assessments, thus a
- **Succession Planning**: AI systems analyze employee performance data and potential, helping organizations ider
- **Employee Engagement**: Chatbots powered by AI can facilitate feedback loops between employees and management,

As AI continues to evolve, HR departments leveraging these agents can realize substantial improvements in both ef

## 5. AI Agents in Finance
The finance sector is seeing extensive integration of AI agents that enhance financial practices:

- **Expense Tracking**: Automated systems manage and monitor expenses, flagging anomalies and offering recommenda
- **Risk Assessment**: AI models assess credit risk and uncover potential fraud by analyzing transaction data and
- **Investment Decisions**: AI agents provide stock predictions and analytics based on historical data and curren

The incorporation of AI agents into finance is fostering a more responsive and risk-aware financial landscape.
```

Ref: <https://docs.crewai.com/en/quickstart>

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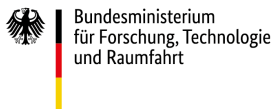
Summary



AutoGen

crewai

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